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Phase-specific determinants of 100 m freestyle performance in elite swimmers

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The 100 m freestyle is a key event in competitive swimming, where marginal gains across race phases—start, clean swim, turn, and finish—are vital for elite performance. However, sex-specific biomechanical contributions remain underexplored, especially in underwater breakout strategies. This study identifies phase-specific performance determinants, emphasizing sex-stratified correlations and practical insights for optimizing elite-level outcomes. High-speed video analysis and official race data from 59 elite swimmers (29 males and 30 females, including Olympic medalists) were utilized. Spatiotemporal variables and stroke efficiency metrics were collected for each phase. Pearson's correlation coefficients assessed relationships between race times and phase-specific variables, while repeated measures ANOVA evaluated stability. All clean swim spatiotemporal variables were highly associated with the final race time, especially the first lap (males: r = 0.806, $p < 0.001$; females: r = 0.850, $p < 0.001$). A robust start (males: r = 0.501, $p = 0.006$; females: r = 0.560, $p = 0.001$) and efficient turn (males: r = 0.588, $p = 0.001$; females: r = 0.592, $p = 0.001$) were also crucial, with underwater variables significantly contributing to a faster start and turn. Sex-specific differences in velocity for turn and finish strategies were identified. The study concludes that elite 100 m freestyle performance depends on explosive starts, sustained clean-swimming velocity, and efficient turns, with sex-specific adaptations in pacing and underwater strategy.

Keywords Performance analysis, Race strategy, Swimming phases

The 100 m freestyle is a premier event in competitive swimming, where even marginal gains can result in world record-breaking performances[1]. The recent achievement by Chinese swimmer Zhanle Pan at the Paris 2024 Olympic Games, setting a new world record of 46.40 s, underscores the significance of incremental improvements over the 100-m course. The swimming community continuously seeks to shave off critical milliseconds through enhanced performance during pivotal race moments[2]. Developing and executing a smart race strategy is essential for success in competitive swimming. Despite this, there is a persistent need to delve deeper into the specific factors within key race phases that contribute to the outcome in a competitive setting. The findings not only offer valuable insights for aspiring athletes but also enhance the competition strategies of elite swimmers. This improvement allows them to break world records in the event, pushing the boundaries of human performance.

Race analysis has emerged as an indispensable tool for coaches and athletes, providing insights into areas for improvement[3]. Recently, practitioners, analysts, and researchers have become increasingly interested in the role played by the start, turn, and finish[4–6]. The start is the time spent between the starting signal and reaching the 15 m mark[7]. The clean swim is the time spent traveling the 30 m in the middle of a long course swimming pool, and the turn is delimited as the time between the last 5 m of a lap and the first 15 m of the next one[8]. The finish is considered as the last 5 m of the race[9]. Different phases of the race have different partial contributions to the final race time. Such as the 100 m freestyle, the start and turn account for nearly a third of the final race time[7]. In Olympic pool swimming, the horizontally oriented jump forward from the elevated block position enables swimmers to kick-start the race with more than twice the actual race velocity and provides a huge initial propulsion for subsequent full-stroke swimming[10,11]. Contrary to Mason et al.[5], our data reveal underwater distance, not reaction time, as the dominant start predictor. A moderate-high correlation was found between the final race time and the start main outcome for the 100 m freestyle[4,9]. Previous studies provide substantial insights on how the fastest swimmers present better clean swimming (i.e. stroke mechanics) in sprinting events[2,12]. This study provides a detailed analysis of specific segments within the 100-m freestyle, with a particular focus on the speed strategy throughout the race. Prior research has predominantly focused on the start and turn phases in

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swimming. However, no existing study has systematically quantified the sex—specific contributions of different phases or the underwater breakout strategies among elite athletes in the 100-m freestyle event. Moreover, comprehensive analyses encompassing all segments of the race are relatively rare. Against this backdrop, the present study aims to identify the key variables in the start, turn, glide, and finish phases that are crucial for achieving faster race times. Additionally, it aims to explore how these variables exert their influence on performance during each distinct phase of the 100-m freestyle race. This study hypothesizes that the critical role of a robust start and efficient turn in enhancing race times, with the underwater phase being the most influential, the speed remains relatively average in the stroke phase.

Methods

Participants

We analyzed the performances of 59 finalists (29 males and 30 females) in the 100 m freestyle event at top-tier Chinese competitions and the 19th Asian Games in Hangzhou. These swimmers, including two Olympic champions and eight medalists, represented the highest level of competitiveness, with their performances closely mirroring world record paces. The mean performance of the males and females corresponded to 94.98% and 95.02% in the 100 m freestyle world records, respectively. This work received ethical approval from the Ethics Committee of Zhejiang Normal University (Ethical Approval Number: ZSRT2024283) and adhered to the principles of the Declaration of Helsinki.

Data collection

The official race times (final race times, every 50 m split time and reaction time) were provided by a flashlight connected to the official timing system, these were retrieved from the official website (https://www.swimming.org.cn/index.html). Two high-definition cameras (SONY ILME-FX6V) recorded each swimmer at 50 frames per second with a shutter speed of 1/1000th of a second. The video delivered real-time recordings using tracking from high-definition fixed-focus cameras. Two fixed video cameras were located on the top row of spectator seats, approximately 6 m above and 7 m away from the side of the pool. The start flashing light was synchronized with the official timer and was visible by all cameras. The start flashing light was used as a reference to set the time-stamp on an in-house customized software for race analysis in competitive swimming. Custom software was utilized for race analysis, with movement coordinates reconstructed using (2D-DLT) algorithms. Distances for start and turn variables were calibrated against pool markers (i.e., 5 m and 15 m marks in the swim lanes)[7,13]. Each start, turn, clean swim, and finish performance was analyzed individually for each swimmer. A single trained evaluator performed all video analyses to ensure consistency, minimizing inter-rater variability.

Start and turn

For the start phase, the analysis focused on the following variables: (i) reaction time, also referred to as block time, which is the interval from the starting signal to the moment the swimmer's feet leave the block; (ii) flight time, the interval from the moment the toes leave the block to when the hands enter the water; (iii) entry time, the interval from the starting signal to when the hands enter the water; (iv) entry distance, the distance from the starting head-wall to where the hands enter the water; (v) underwater time, the interval from when the hands enter the water to when the head breaks through the water surface; (vi) underwater distance, the distance from where the hands enter the water to when the head breaks through the water surface; (vii) underwater speed, calculated between the entry time and the moment the head breaks the water surface; (viii) water breakout time, the interval from the starting signal to when the head breaks through the water surface; (ix) water breakout distance, the distance from the starting head-wall to the point where the head breaks the water surface; and (x) the 15 m mark time, the interval from the starting signal to when the swimmer's head reaches the 15 m mark. The 15 m mark time was selected as the primary outcome measure for the start phase[7,14,15].

For the turn phase, the variables under scrutiny included: (i) the 5 m-in time, the interval from reaching the 45 m mark to touching the wall, as recorded by the official split time; (ii) water break time, the interval from touching the wall to when the head breaks through the water surface; (iii) water break distance, the distance from the wall to where the head breaks through the water surface; (iv) underwater speed during the turn, calculated from the moment of touching the wall to when the head breaks the water surface; (v) the 15 m-out time, the interval from touching the wall to reaching the 15 m mark of the next lap; and (vi) the total turn time, the interval from reaching the 45 m mark to the 15 m mark of the following split. The total turn time was designated as the primary outcome for the turn phase[7,13,15].

The clean swim and finish

In the context of two laps, clean swim performance was defined as the time spent covering the distance between the 15th and 45th meters and between the 65th and 95th meters, respectively[13]. The clean swim speed during the intermediate 30 m was also calculated to facilitate a comparison with the finish speed. For the 100 m event, both laps were analyzed to mitigate any potential advantages gained from the previous turn and to account for spatial–temporal adjustments for the next turn. In each lap, the following pace variables were examined: (i) the clean swim performance (s); (ii) the clean swim speed (v, m/s) ; (iii) the stroke frequency (SF, cycles/min) is the number of complete a cycle stroke (i.e., one hand entry to other hand entry) taken each minute; (iv) the stroke length (SL, m/cycles) is the distance a swimmer swims in a cycle stroke, and (v) the stroke index (SI, m^2/s) (describes a swimmer's ability to move with a minimum number of strokes at a given speed). Afterward, the mean values of all complete strokes during the middle 30 m were utilized for analysis. The v was calculated as:$v = d/t$, where d is the distance and t the time swum. The SF was assessed in the race analysis software. It was obtained by computing the period of the time spent to complete a full stroke cycle (during the intermediate

30 m), having as a reference always the hand closest to the camera. The SL was calculated as $SL = v/SF$[16], and the SI as $SI = v \cdot SL$ [17]. The finish was considered as the last 5 m[9]. Therefore, it was assessed as the time spent to travel the last 5m of the race, and the corresponding speed ($v = d/t$)[8,15].

Define the way of phases

For analysis, races were divided into four phases: start, turn, finish, and first and second half of the stroke phase. The distance of starting 15 m (S_{15}), 1st half of the stroke phase (HS) (i.e., the distance between reaching the 15 m mark and the 45 m mark), T_{20} (i.e., the distance between reaching the 45 m mark to the touch on the wall and reaching the 65 m mark), 2nd HS (i.e., the distance between reaching the 65 m mark and the 95 m mark) and the final 5 m of the race (F_5) are chosen to symbolize the four-phase during each 100 m swimming journey (Fig. 1). Velocity at the Start 15 m (VS_{15}), V1st HS, Velocity during the 20 m Turn (VT_{20}), V2nd HS, and Velocity in the Final 5 m Sprint (VF_5) represent the average speed of each corresponding section of the swimming journey. In addition, the V1st HS was divided into V_{15-25} and V_{25-45}, and the V2nd HS was divided into V_{65-75} and V_{75-95}. Similarly, TS_{15}, T1st HS, TT_{20}, T2nd HS, and TF_5, respectively, represent the time performance of each corresponding section of the swimming journey.

Data analysis

All the data analysis was conducted with SPSS 22.0 software. The Shapiro–Wilk and Levene tests were used to assess the normality and homoscedasticity assumptions, respectively. The mean ± one standard deviation (SD) was computed for all variables.

Pearson's correlation coefficient was used to assess the 100 m race times with start, turn, clean swim, and finish phases time variables respectively. As rule of thumb, for qualitative and effect size assessments the correlation/ relationship was defined as: (1) without effect if $0 < r \leq 0.20$; (2) minimum if $0.20 < r \leq 0.40$; (3) moderate if $0.40 < r \leq 0.60$; (4) strong if $0.60 < r \leq 0.80$ and (5) very strong if $r > 0.80$[19].

The swimmers' stability was assessed based on (1) mean stability and (2) normative stability[18]. Mean stability refers to the persistence of the magnitude of change over the race of a given variable. This was assessed by the ANOVA repeated measures, followed by the Bonferroni post-hoc test to verify significant differences between each pairwise ($p < 0.05$). The effect size (eta square $-\eta^2$) was computed and interpreted as: (1) without effect if $0 < -\eta^2 \leq 0.04$; (2) minimum if $0.04 < \eta^2 \leq 0.25$; (3) moderate if $0.25 < \eta^2 \leq 0.64$ and; (4) strong if $\eta^2 > 0.64$[19]. The coefficient of variation (CV, in %) was calculated for each pairwise. Cohen's d was selected as standardized effect size for each pairwise, and interpreted as: (1) small effect size $0 \leq |d| \leq 0.2$; (2) medium effect size if $0.2 < |d| \leq 0.5$ and; (3) large effect size if $|d| > 0.5$[20]. The variables of the clean swim phase were compared and analyzed by a non-parametric test.

Results

Tables 1 and 2 present the descriptive statistics and Pearson's correlation coefficient between the 100 m performance with the start, clean swim, turn, and finish variables. The 100 m performance presented the strongest correlation (male: $r = 0.806$, $p < 0.001$; female: $r = 0.850$, $p < 0.001$) with T1st HS. A shorter T1st HS generally corresponds to better clean swim performance, indicating that swimmers who can maintain higher speed throughout the first half of the race tend to perform better. Overall, the 100 m race time showed significant and moderate to strong associations with the clean swim and a significant and moderate association with other phases' variables (Tables 1 and 2). For males, swimmers took 5.64 ± 0.18 s to achieve the 15 m mark (start main outcome) and 9.72 ± 0.23 s during the total turn (turn main outcome). The start accounted for 11.58 ± 0.33% of the total race time, the clean swim 63.30 ± 0.58%, the turn 19.94 ± 0.38%, and the finish 5.18 ± 0.30% (Table 1). For females, more times were spent than males in the S_{15} (6.41 ± 0.17 s) and T_{20} (10.90 ± 0.30 s). In female swimmers, the relative contribution showed the same trend as in males. The start accounted for 11.82 ± 0.26% of

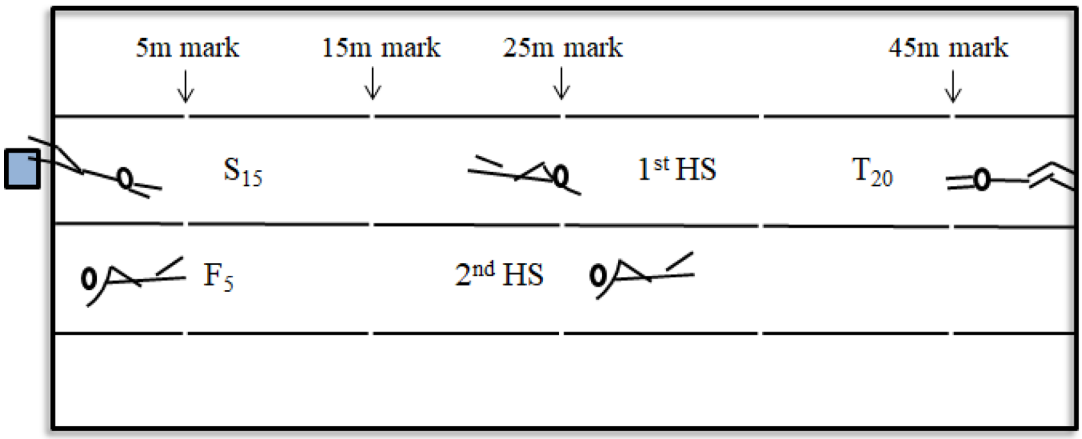

Fig. 1. The map illustrates the marking of the 50 m pool section.

Variables	Descriptive Mean ± SD	Relative contribution (%)	r	*p*
Performance (s)	48.73 ± 0.83			
Start				
TS_{15} (s)	5.64 ± 0.18	11.58 ± 0.33	0.501**	0.006
Reaction time (s)	0.61 ± 0.03		0.414#	0.025
Flight time (s)	0.28 ± 0.05		0.437*	0.018
Entry time (s)	0.90 ± 0.05		0.435*	0.018
Entry distance (m)	3.59 ± 0.26			
Underwater time (s)	3.35 ± 0.41			
Underwater distance (m)	8.29 ± 0.95		−0.427#	0.021
Underwater speed (m/s)	2.49 ± 0.15			
Water breakout time (s)	4.24 ± 0.42			
Water breakout distance (m)	11.88 ± 0.97		−0.420#	0.023
Clean swim				
T1st HS (s)	15.01 ± 0.34	30.79 ± 0.41	0.806**	< 0.001
Swim speed (m/s)	2.00 ± 0.05		−0.808**	< 0.001
Stroke frequency (cycles/min)	49.86 ± 3.28		0.376#	0.045
Stroke length (m/cycle)	2.42 ± 0.15			
Stroke index (m^2/s)	4.83 ± 0.32		−0.368#	0.049
T2nd HS (s)	15.84 ± 0.40	32.51 ± 0.53	0.778**	< 0.001
Swim speed (m/s)	1.90 ± 0.05		−0.783**	< 0.001
Stroke frequency (cycles/min)	48.41 ± 3.54			
Stroke length (m/cycle)	2.36 ± 0.16			
Stroke index (m^2/s)	4.47 ± 0.34		−0.439#	0.017
Turn				
TT_{20} (s)	9.72 ± 0.23	19.94 ± 0.38	0.588**	0.001
5 m-in (s)	2.84 ± 0.11		0.809##	< 0.001
Water breakout time (s)	2.75 ± 0.45			
Water breakout distance (m)	7.11 ± 0.88		−0.370*	0.048
Underwater speed (m/s)	2.60 ± 0.16		−0.426#	0.021
15 m-out (s)	6.87 ± 0.15		0.912##	< 0.001
Finish				
TF_5 (s)	2.52 ± 0.14	5.18 ± 0.30		
VF_5 (m/s)	1.99 ± 0.11			

Table 1. Pearson's correlation coefficient between the male's 100 m performance and the start, clean swim, turn and finish variables. *Represents the moderate effect of 100 m performance; **represents the strong effect of 100 m performance; # represents the moderate effect of phase variables; ## represents the strong effect of phase variables.

the total race time, the clean swim 63.03 ± 0.55%, the turn 20.09 ± 0.44%, and the finish 5.07 ± 0.17% (Table 2). There was no significant difference in the relative contribution of different genders ($p > 0.05$).

Regarding the start, a significant and moderate-strong race effect was verified in both males and females in the 15 m start time. For males, reaction time (RT) (0.61 ± 0.03 s) was found to have a moderate effect on the S_{15} performance of the swimmers $(\mathrm{r} = 0.501, p = 0.006)$. Underwater distance (UD) and water breakout distance (WBD) variables demonstrated a moderate negative correlation on the start performance $(\mathrm{UD}: \mathrm{r} = -0.427, \mathrm{p} = 0.021; \mathrm{WBD}: \mathrm{r} = -0.420, p = 0.023)$. Males covered a longer WBD (8.29 ± 0.95 m) compared to females (7.86 ± 0.93 m). The 100 m performance presented a moderate correlation with flight (0.28 ± 0.05 s) and entry time (0.90 ± 0.05 s). While for females, only the underwater speed (US) (2.27 ± 0.11 m/s) variable demonstrated a moderate negative correlation on the start performance $(\mathrm{r} = -0.486, p = 0.006)$.Of course, the overall average speed in the departure phase is the fastest of all phases (male: 2.661 ± 0.09 m/s; female: 2.340 ± 0.06 m/s).

As for the clean swim, T2nd HS showed a relatively strong correlation with the 100 m performance besides T1st HS (male: $\mathrm{r} = 0.778, p < 0.001$; female: $\mathrm{r} = 0.641, p < 0.001$). The data shows that SF, SL, and SI of 1st HS significantly higher than SF, SL and SI of 2nd HS for male (Z (SF) = 3.406, $p = 0.001$; Z (SL) = 2.952, $p = 0.003$; Z(SI) = 4.681, $p < 0.001$). But for females, SF and SI represent a significant difference (Z (SF) = 4.741, $p < 0.001$; Z(SI) = 4.638, $p < 0.001$). As for SL, there was no significant difference (Z (SL) = 0.314, $p = 0.754$). The decrease of swimming velocity was approximately 5% between the 1st HS and 2nd HS in the full-exertion 100-m front crawl. Figure 2 presents a distribution of speeds at different points during the race, indicating variations in pacing

Variables	Descriptive Mean ± SD	Relative contribution (%)	r	*p*
Performance (s)	54.28 ± 0.90			
Start				
TS_{15} (s)	6.41 ± 0.17	11.82 ± 0.26	0.560**	0.001
Reaction time (s)	0.66 ± 0.03			
Flight time (s)	0.28 ± 0.04			
Entry time (s)	0.95 ± 0.05			
Entry distance (m)	3.22 ± 0.18			
Underwater time (s)	3.47 ± 0.44			
Underwater distance (m)	7.86 ± 0.93			
Underwater speed (m/s)	2.27 ± 0.11		−0.486##	0.006
Water breakout time (s)	4.42 ± 0.45			
Water breakout distance (m)	11.09 ± 0.88			
Clean swim				
T1st HS (s)	16.46 ± 0.43	30.32 ± 0.45	0.850**	< 0.001
Swim speed (m/s)	1.82 ± 0.05		−0.855**	< 0.001
Stroke frequency (cycles/min)	50.54 ± 2.75			
Stroke length (m/cycle)	2.17 ± 0.12		−0.404 #	0.027
Stroke index (m^2/s)	3.96 ± 0.28		−0.689##	< 0.001
T2nd HS (s)	17.75 ± 0.29	32.71 ± 0.44	0.641**	< 0.001
Swim speed (m/s)	1.69 ± 0.03		−0.673**	< 0.001
Stroke frequency (cycles/min)	46.94 ± 2.31		−0.373#	0.042
Stroke length (m/cycle)	2.16 ± 0.10			
Stroke index (m^2/s)	3.66 ± 0.17			
Turn				
TT_{20} (s)	10.90 ± 0.30	20.09 ± 0.44	0.592**	0.001
5 m-in (s)	3.20 ± 0.16		0.846##	< 0.001
Water breakout time (s)	2.40 ± 0.67			
Water breakout distance (m)	5.82 ± 1.21			
Underwater speed (m/s)	2.49 ± 0.28			
15 m-out (s)	7.71 ± 0.18		0.883##	< 0.001
Finish				
TF_5 (s)	2.75 ± 0.11	5.07 ± 0.17	0.603**	< 0.001
VF_5 (m/s)	1.83 ± 0.08		−0.671**	< 0.001

Table 2. Pearson's correlation coefficient between the female's 100 m performance and the start, clean swim, turn and finish variables. *Represents the moderate effect of 100 m performance; ** represents the strong effect of 100 m performance; # represents the moderate effect of phase variables; ## represents the strong effect of phase variables.

strategies among swimmers. Table 3 highlights related to the performance in 100 m freestyle swimming correlation with various velocity variables. The data indicates that the male 100 m performance presented the strongest correlation with V_{65-75} ($r = -0.818, p < 0.001$) and V_{25-45} ($r = -0.790, p < 0.001$). And the female 100 m performance presented the strongest correlation with V_{15-25} ($r = -0.884, p < 0.001$). The swimmers measured the results of the mean stability of the clean swim velocity during the 100 m race, greenhouse-geissler's estimate of the sphericity deviation is $\omega = 0.465$, V_{15-25}, V_{25-45}, V_{65-75}, V_{75-95} and VF_5 have a significant difference ($\text{male}: F(1.86, 52.13) = 36.98, p < 0.001, \eta^2 = 0.57;\ \text{female}: F(2.48, 71.90) = 98.57, p < 0.001, \eta^2 = 0.77$). Table 4 shows the results of the multiple comparisons, the results showed that the average velocity of the different sexes was V_{15-25}, VF_5, V_{25-45}, V_{65-75} and V_{75-95} in order from fast to slow. But for males there is no significant difference between VF_5 and V_{25-45} (t = 0.82, p = 1.000, d = 0.17), VF_5 and V_{65-75} (t = 2.13, p = 0.369, d = 0.45). For females, there is no significant difference between VF_5 and V_{25-45} (t = 2.13, p = 0.448, d = 0.39).

In the turn phase, the 5 m—in, the 15 m–out, Water Breakout Distance (WBD) and Underwater Speed (US). These variables also showed a significant impact on the turn performance of the swimmers, with males having a mean underwater speed of 2.60 ± 0.16 m/s and females slightly faster at 2.49 ± 0.28 m/s; males had a mean WBD of 7.11 ± 0.88 m and females slightly longer at 5.82 ± 1.21 m. Combining the results from Table 3 and Fig. 2, the average velocity for the turn phase exceeds 1st HS, whereas the females' turn phase falls below 1st HS. But the correlation between females and 100 m final performance was higher than for males.

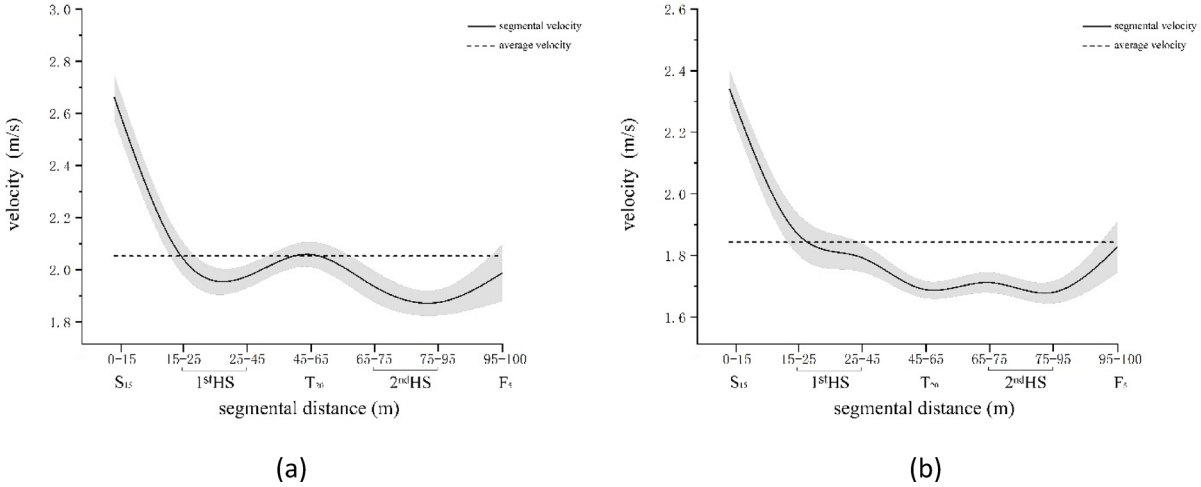

Fig. 2. A linear relationship between segmented distance and average velocity. **a** For males; **b** For females.

Variables	Descriptive		r		p	
	Mean ± SD					
	Male	Female	Male	Female	Male	Female
Performance (s)	48.73 ± 0.83	54.28 ± 0.90				
VS_{15}	2.661 ± 0.09	2.340 ± 0.06	− 0.510*	− 0.560*	0.005	0.001
V1st HS						
V_{15-25}	2.063 ± 0.06	1.882 ± 0.07	− 0.628*	− 0.884*	< 0.001	< 0.001
V_{25-45}	1.970 ± 0.04	1.796 ± 0.05	− 0.790*	− 0.700*	< 0.001	< 0.001
VT_{20}	2.059 ± 0.05	1.690 ± 0.03	− 0.596*	− 0.673*	0.001	< 0.001
V2nd HS						
V_{65-75}	1.938 ± 0.06	1.712 ± 0.03	− 0.818*	− 0.506*	< 0.001	0.002
V_{75-95}	1.874 ± 0.05	1.680 ± 0.04	− 0.665*	− 0.598*	< 0.001	< 0.001
VF_5	1.988 ± 0.11	1.828 ± 0.08		− 0.671*		< 0.001

Table 3. Pearson's correlation coefficient between the 100 m performance and velocity variables. *Represents the significant difference.

The velocity curve in Fig. 2 demonstrates a noticeable upward trend in the average velocity during the finish phase. Only females' finish variables had a strong correlation with 100 m performance (Time and velocity: $p < 0.001$).

Discussion and implications

This study has characterized the impact of various phases on 100 m freestyle performance, revealing that each phase significantly influences final race times. A strong start, efficient turns, and maintaining velocity throughout the race are crucial for elite swimmers. Targeted training in these areas can lead to significant improvements in race times. However, the gender differences observed in the study warrant further exploration to provide a more comprehensive understanding of the underlying factors contributing to these disparities.

Start

The start phase is critical in determining the initial momentum and velocity that swimmers carry into the race. The study found that a robust start, characterized by a fast 15 m mark time, significantly correlates with better overall race performance. Swimmers who took more time to reach the 15 m mark also took longer to complete the race. In the 100 m events, the 15 m start time may account for between 11 and 12% of the final race time in both sexes[7]. The contribution of start performance to 100 m Freestyle races did not differ from previously reported findings of start performance contribution to long-course races (11.96 ± 0.26%)[7].

Firstly, the negative correlations between performance times and early-stage velocities ($V1^{st}HS$, V_{15-25}) underscore the importance of a strong start and maintaining high speed in the initial meters of the race. In 100 m events, swimmers reaching faster underwater speeds had a meaningful advantage on swimming start, and consequently on the final race times[21,22]. Coaches and athletes should focus on developing explosive power and technique to maximize speed during these critical phases. Studies noted that the faster swimmers achieving the 15 m mark also reached faster velocities at the beginning of the clean swim phase[5,11]. This suggests that faster

male / female	V_{15-25}	V_{25-45}	V_{65-75}	V_{75-95}	VF_5
V_{15-25}		MD=0.093* P<0.001	MD=0.125* P<0.001	MD=0.189* P<0.001	MD=0.075* P=0.019
V_{25-45}	MD=-0.086* P<0.001		MD=0.032* P=0.003	MD=0.096* P<0.001	MD=-0.018 P=1.000
V_{65-75}	MD=-0.170* P<0.001	MD=-0.084* P<0.001		MD=0.064* P<0.001	MD=-0.049 P=0.369
V_{75-95}	MD=-0.202* P<0.001	MD=-0.116* P<0.001	MD=-0.031* P=0.006		MD=-0.114* P<0.001
VF_5	MD=-0.054* P=0.002	MD=0.032 P=0.448	MD=0.116* P<0.001	MD=0.148* P<0.001	

Table 4. Results of pairwise comparison of velocity in different phases. *MD* mean difference. *Represents significant difference.

starters are also able to maintain their speed during the clean swimming phase. Indeed, our data showed that the first lap was the fastest, and this must be related to fast starts. A swimmer should aim to maintain the initial speed in the stroke phase by acquiring a technique that minimizes deceleration at each phase before stroking.

Secondly, studies indicated that in 100 m events, swimmers reaching further underwater distance had a meaningful advantage in swimming start, and consequently on the final race times[21,22]. Relatively longer WBD suggests better starts, indicating that swimmers who break out water farther tend to have an advantage in this phase.

Moreover, they should be advised to understand the critical moment when the underwater speed is lower than the swim speed that they should achieve in the surface phase[23]. Previous studies have shown that the average velocity is approximately 1.92 m/s during the 7.5–10 m segment, and the average velocity is approximately 1.85 m/s during the 10–15 m segment[15]. The results of this study showed that the average velocity of the first clean swimming phase (i.e., the 15–25 m) after the start was 2.06 m/s for males and 1.88 m/s for females. Once the swimmer decreases his forward speed to about 1.75 to 2.0 m/s[24](from 3 to 3.5 m/s in which the wall push-off is performed) then he/she must finalize the gliding phase and begin with the underwater propulsion phase, to reduce the swimmer's deceleration[25–29].

In summary, these findings suggest that while both genders benefit from a strong start, the specific determinants of start performance differ between males and females. The observed differences in start performance between males and females can be attributed to several factors. Firstly, males exhibited a longer water breakout distance (WBD) compared to females, indicating that they were able to maintain higher underwater speeds for a longer duration. This may be due to greater lower limb explosive power and a more streamlined body position during the underwater phase, which reduces drag and enhances propulsion. Additionally, the higher underwater speed (US) observed in males compared to females suggests that males may have a greater ability to generate force during the initial push-off from the starting block. This could be related to differences in muscle mass, strength, and neuromuscular activation patterns between the genders. Coaches should refer to the male and female athletes' lower limb explosive power and speed at 10 m, and breaking water in the appropriate place may be more conducive to the improvement of the starting phase and overall performance. Future research should investigate the specific biomechanical and physiological factors that contribute to these gender differences in start performance.

The clean swim

The clean swim phase emerged as the most influential phase, accounting for over 63% of the total race time. Swimming research historically puts a lot of focus on the clean swim phase because this phase is highly associated with the final race time[7,30]. Our data showed that all clean swim spatiotemporal variables (i.e., lap #1 and lap #2) had a positive and significant relationship with the final race time. For both males and females, the first lap (15–45 m) showed the strongest correlation with final race time, indicating that maintaining high speed in the initial segment of the race is crucial.

This has also been confirmed in previous studies, a 5.7% decrease in swimming velocity during the second half of a long-course 100-m race[31]. The decrease in swimming velocity observed in the current study was slightly smaller than those found in the previous research. In all cases, the swimming velocity on the start emersion (first lap) was much faster than on the turn emersion (second lap) (for males and females)[32]. The initial segment of the second lap (i.e., the 65–75 m) had the strongest relationship with the final time in all 100 m events. In other words, maintaining a relatively high speed after the turn phase positively impacts final race performance. Due to the impact of fatigue, it was possible to note that 100 m swimmers slightly slowed down in the last lap, between the intermediate 30 m; this is consistent with the results of previous studies. This was also observed in elite swimmers across different distances and swimming strokes[30].

Since the hand is considered the primary source of propulsive force, the decreased swimming velocity likely resulted from the reduced hand velocity[31]. The authors suggested that improved clean swim performance reflects an effective skill that sustains a high stroking speed. The data indicate that while maintaining speed throughout the race is crucial, there may be a point of diminishing returns where additional effort does not significantly enhance performance. This is evident from the slightly lower but still significant correlations seen in later segments (e.g., V_{75-95}). Athletes might benefit from pacing strategies that prioritize energy conservation after the initial burst, ensuring they have enough stamina to maintain a competitive speed towards the end of the race.

The gender differences observed in clean swim performance can be attributed to several factors. Firstly, males exhibited higher SF and SL in the first lap compared to females, which contributed to their higher swimming velocities. This suggests that males may have a greater ability to generate propulsive force with each stroke, possibly due to differences in muscle strength and coordination. Additionally, the higher stroke index observed in males indicates that they were more efficient in converting their stroke mechanics into forward propulsion. These differences may be related to physiological factors such as muscle mass, strength, and endurance, as well as biomechanical factors such as stroke technique and body position in the water. Future research should explore the specific training interventions and technical adjustments that can help females optimize their clean swim performance to bridge the gap with males.

Turn

Combined with the data results, sex-specific differences were noted, with male swimmers exhibiting higher average speeds during the turn phase; males' average speed was 21.9% higher than females. From a neuromuscular perspective, suggesting a potential strength advantage, both start and turn performances require a high force production from the starting block or pool wall[33,34]. After wall contact, an efficient underwater phase requires a strong leg push and a short gliding phase until the velocity has dropped to underwater kicking velocity[32,35,36]. This may be related to the absolute strength of the lower limbs of different genders. Female swimmers probably traveled 1–2 m shorter underwater distances with lower underwater velocities[37,38], and the swimmers performed a short propulsion, thus reducing their gliding distance for faster entry into the freestyle phase[39]. swimmers aimed to reduce their distance so as not to increase glide time, drag, and energy expenditure and thereby cause a loss of speed; this could be a strategy in a short-distance event like 100 m. Additionally, the shorter underwater distances observed in females may be due to a strategy of minimizing glide time to reduce drag and energy expenditure, thereby allowing for a quicker transition to the surface swim stroke. This strategy may be more effective for females, given their relatively lower underwater speeds. As for the start, both male and female swimmers break the water sooner in Freestyle, and as such, resume the swim stroke also sooner[7]. Thus, knowing the optimal moment to transition from the underwater to the surface stroke is crucial for swimmers aiming to enhance their start performance[40].

Faster 5 m in times for Freestyle tumble turns result as a consequence of faster swimming velocities when approaching the wall[11], and the strong positive correlation of this phase fully shows that it plays a pivotal role in the whole process of the turn. In the turn, the push-off time and peak force on the wall performing the flip turn, and the wall contact time (hands and feet) and a quick pullout in the open turn may be the key determinants[41,42]. Due to the conditions of the competition, these data are not available. This would be a limitation of the study. Based on the available results, we can still offer swimmers a choice of two main strategies: (i) increase their underwater break distance/time to save energy (underwater profile); or (ii) start the swim stroke sooner, and hence breaking the water sooner as well (surface profile)[10].

The final

The finish phase is the final stretch of the race where swimmers aim to maintain or increase their speed to secure a better position. In the 100 m events, male and female swimmers increased the swim speed in the last 5 m (in comparison to the last lap intermediate 30 m), the finish phase's impact on overall performance was more pronounced in female swimmers, emphasizing the need for strategies to maintain or increase speed in the final stretch. However, maintaining or increasing speed in the final stretch may be more critical for female swimmers. This could be due to differences in fatigue resistance and energy management strategies between the genders. The finish is related to the ability to keep or increase the clean swim speed in the last 5 m stretch[2,9]. We contend that the effects of fatigue are comparatively minor, given that the speed can reach the velocity characteristic of the first stroke phase (V_{15-25})[15]. Future research should explore the specific physiological and biomechanical factors that contribute to these gender differences in finish performance, as well as investigate training interventions that can enhance finish speed in both genders.

Limitations

This study still has certain limitations. To minimize the differences between evaluators, a trained evaluator with 5 years of analysis experience was specifically assigned to handle all video analysis work. However, this study did

not conduct an in-depth investigation into the inter-rater reliability during the manual video analysis process. In future studies, we will focus on improving this aspect to further enhance the rigor and reliability of the research.

Conclusion

This study provides a comprehensive analysis of the phase-specific determinants of 100 m freestyle performance in elite swimmers, offering novel insights into the distinct contributions of the start, clean swim, turn, and finish phases. Through the integration of high-speed video analysis and sex-stratified correlations, we identified key variables that significantly impact race outcomes, thereby addressing gaps in prior literature. Our findings underscore the critical role of a robust start, characterized by optimal reaction time and underwater distance, in establishing initial momentum. The clean swim phase emerged as the most influential, accounting for over 63% of the total race time, with maintaining high velocity throughout this phase being essential for superior performance. The turn phase also demonstrated a significant impact, with underwater speed and breakout distance being crucial for a quick transition back to swimming. Notably, the finish phase, particularly for female swimmers, exhibited a strong correlation with final race performance, highlighting the importance of accelerating in the final stretch.

The study's specific contributions include the identification of sex-specific differences in performance determinants, which have not been systematically explored in previous research. For instance, males exhibited higher underwater speeds and longer water breakout distances during the start and turn phases, suggesting greater lower limb explosive power and neuromuscular efficiency. In contrast, females showed a stronger correlation between finish variables and overall performance, indicating potential differences in fatigue resistance and energy management strategies. These findings provide actionable insights for coaches and athletes, emphasizing the need for targeted training programs that address these gender-specific factors.

Moreover, this study highlights the importance of pacing strategies in maintaining high velocity throughout the race. The data indicate that while maintaining speed is crucial, there may be a point of diminishing returns where increased effort does not significantly improve performance. This suggests that athletes may benefit from pacing strategies that prioritize energy conservation after the initial burst, ensuring they have enough stamina to maintain a competitive speed toward the end of the race.

In conclusion, our results emphasize the multifaceted nature of 100 m freestyle performance, with each phase contributing uniquely to the final race outcome. By focusing on enhancing explosive power, optimizing stroke efficiency, and implementing effective pacing strategies, athletes can potentially achieve marginal gains that make a significant difference at the elite level of competition. Future research should explore the integration of these phase-specific strategies into comprehensive training programs and investigate their impact on performance across a range of race distances. Additionally, further exploration of the underlying biomechanical and physiological factors contributing to gender differences in each phase will provide a more comprehensive understanding of the determinants of 100 m freestyle performance.

Data availability

The Data will be made available upon request by contacting Houwei Zhu.

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Author contributions
Q.G. and Z.H.: Conceptualization; X.J. and Z.H.: methodology; X.J.: soft-ware; Z.H. and G.L.: validation; Q.G. and Z.H.: resources; Q.G. and Z.H.: writing—original draft preparation; H.Z. and Q.G.: writing—review and editing; H.Z.: supervision; H.Z.: project ad-ministration; H.Z.: funding acquisition. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Human Ethics
Its contents and methods have been determined to be legally valid and approved by the Ethics Committee of the Zhejiang Normal University (Ethical Approval Number: ZSRT2024283). The need for informed consent was waived by the Ethics Committee of the Zhejiang Normal University because all video data are from open competitions without intervention, and the video capture method used did not affect the athletes' performance in any way. Our research adhered to the principles of the Declaration of Helsinki.

Informed consent

Informed consent was obtained from all subjects involved in the study.

Additional information

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